

Homework 5

Homotopy Invariance

Math 615: Algebraic Topology I
Fall 2026 | September 28 and 30

Use Lecture 5. Give proofs, state the source and target of every map you define, and keep the signs in chain calculations explicit. Coefficients are integers. Write $I = [0, 1]$ and use all singular simplices, including degenerate ones.

A homotopy from $f : X \rightarrow Y$ to $g : X \rightarrow Y$ is a continuous map $h : X \times I \rightarrow Y$ with $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$. A homotopy inverse of f is a map $g : Y \rightarrow X$ with $gf \simeq \text{id}_X$ and $fg \simeq \text{id}_Y$. A nonempty space is contractible if its identity map is homotopic to a constant map.

The singular group $C_n(X)$ is free abelian on continuous maps $\sigma : |\Delta^n| \rightarrow X$. Write σ also for its generator. Here

$$|\Delta^n| = \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum_i t_i = 1\},$$

with standard vertices $e_0, \dots, e_n$. The affine map $d^i : |\Delta^{n-1}| \rightarrow |\Delta^n|$ inserts a zero in position i , and $\partial\sigma = \sum_i (-1)^i \sigma d^i$ for $n > 0$. For a subspace $X' \subseteq X$, put $C_\bullet(X, X') = C_\bullet(X)/C_\bullet(X')$ and $H_n(X, X') = H_n(C_\bullet(X, X'))$.

Exercise 1. The Homotopy Category and Homotopy Invariant Functors

The category hTop has spaces as objects and homotopy classes of continuous maps as morphisms. The quotient functor is $q : \text{Top} \rightarrow \text{hTop}$. For a category C , a functor $F : \text{Top} \rightarrow C$ is homotopy invariant if $F(f) = F(g)$ whenever $f \simeq g$.

1. Prove that composition in hTop is well defined. Prove that F is homotopy invariant if and only if there is a unique functor $\bar{F} : \text{hTop} \rightarrow C$ with $F = \bar{F}q$.
2. Prove that $[f]$ is an isomorphism in hTop if and only if f is a homotopy equivalence. Deduce that every homotopy invariant functor sends homotopy equivalences to isomorphisms, and identify the inverse of $F(f)$.
3. For $\epsilon \in \{0, 1\}$ define $j_\epsilon : X \rightarrow X \times I$ by $j_\epsilon(x) = (x, \epsilon)$, and let $p : X \times I \rightarrow X$ be projection. Show that F is homotopy invariant if and only if $F(p)$ is an isomorphism for every X . In the reverse implication, first show that $F(j_0) = F(j_1)$.

Exercise 2. The Prism over a Singular Edge

Let $h : X \times I \rightarrow Y$ be a homotopy from f to g , and let $\sigma : |\Delta^1| \rightarrow X$ be a singular edge. The affine maps $\tau_0, \tau_1 : |\Delta^2| \rightarrow |\Delta^1| \times I$ have ordered vertices

$$\tau_0 : [(e_0, 0), (e_0, 1), (e_1, 1)], \quad \tau_1 : [(e_0, 0), (e_1, 0), (e_1, 1)].$$

Define $P_1 : C_1(X) \rightarrow C_2(Y)$ on generators by

$$P_1(\sigma) = h(\sigma \times \text{id}_I)\tau_0 - h(\sigma \times \text{id}_I)\tau_1.$$

For a point $x \in X$, let $P_0(x) : |\Delta^1| \rightarrow Y$ be the path $P_0(x)(t_0, t_1) = h(x, t_1)$, extended linearly to $P_0 : C_0(X) \rightarrow C_1(Y)$.

1. Draw the square and its diagonal. Write both affine maps in coordinates (t_0, t_1, t_2) , including their two target coordinates in $|\Delta^1| \times I$.
2. List the three ordered edges of each triangle, with their coefficients in $\partial P_1(\sigma)$. Identify the top, bottom, side, and interior edges. Explain geometrically why the diagonal is interior and the two vertical edges are side edges.
3. Verify directly that

$$\partial P_1(\sigma) + P_0(\partial \sigma) = g_{\#}(\sigma) - f_{\#}(\sigma).$$

In particular, exhibit the two equal singular edges on the diagonal and their opposite coefficients.

Exercise 3. Explicit Homotopies and Homology

A nonempty subset $X \subseteq \mathbb{R}^m$ is star shaped about $p \in X$ if $(1-t)x + tp \in X$ for every $x \in X$ and $t \in I$.

1. Construct a contraction of a star shaped space. Compute all its singular homology groups, including degree zero. Explain why the same argument applies to every nonempty convex subset.
2. For $m \geq 1$, define $r : \mathbb{R}^m \setminus \{0\} \rightarrow S^{m-1}$ by $r(x) = x/\|x\|$, and let $i : S^{m-1} \hookrightarrow \mathbb{R}^m \setminus \{0\}$ be inclusion. Construct a homotopy from id to ir that fixes the sphere throughout. Deduce that i_* and r_* are inverse in every degree. No computation of the homology of a positive-dimensional sphere is required.
3. Take $m = 1$ and compute every homology group of $\mathbb{R} \setminus \{0\}$. Explain why the punctured line cannot be homotopy equivalent to the line.

Exercise 4. Retracts and Chain Homotopies

A retraction of a subspace $X' \subseteq X$ is a map $r : X \rightarrow X'$ with $ri = \text{id}_{X'}$, where $i : X' \hookrightarrow X$ is inclusion.

1. Prove that $i_* : H_n(X') \rightarrow H_n(X)$ is injective and r_* is surjective. If X is contractible and X' is nonempty, use a contraction of X to construct a contraction of X' .
2. Let $u, v : A_{\bullet} \rightarrow B_{\bullet}$ be chain maps and let $K_n : A_n \rightarrow B_{n+1}$ satisfy $v - u = d_B K + K d_A$. Show explicitly that u and v induce the same map on homology.
3. Suppose $u : A_{\bullet} \rightarrow B_{\bullet}$ has a chain map $w : B_{\bullet} \rightarrow A_{\bullet}$ such that wu and uw are chain homotopic to the respective identities. Identify the inverse of u_* , and explain how the prism construction applies this algebraic fact to a homotopy equivalence of spaces.

Exercise 5. The Interval and Its Endpoints

Let $X = I$ and $X' = \{0, 1\}$. Use the long exact sequence associated to

$$0 \rightarrow C_{\bullet}(X') \rightarrow C_{\bullet}(X) \rightarrow C_{\bullet}(X, X') \rightarrow 0.$$

1. Compute $H_n(X')$ and $H_n(X)$ for all n . Identify the map $H_0(X') \rightarrow H_0(X)$ under the identifications with $\mathbb{Z}^2$ and $\mathbb{Z}$.
2. Deduce all the relative groups $H_n(X, X')$. For the singular edge $\sigma : |\Delta^1| \rightarrow I$ given by $\sigma(t_0, t_1) = t_1$, compute its relative class and its image under the connecting map to $H_0(X')$. Distinguish the two point generators from integer coefficients.

3. The reflection $f : I \rightarrow I, f(x) = 1 - x$, preserves X' . Compute its induced map on $H_1(X, X')$. Show that f is homotopic to id_I as a map of spaces, but that there is no such homotopy through maps preserving X' at every time. Explain why there is no contradiction with relative homotopy invariance.

Exercise 6. A Contractible Space with Several Marked Points

Let X be a contractible space and let $X' = \{x_0, \dots, x_{k-1}\} \subseteq X$ be a discrete subspace of $k \geq 1$ distinct points.

1. Compute all the groups $H_n(X, X')$. Identify $H_1(X, X')$ canonically with the kernel of the homomorphism $\mathbb{Z}^k \rightarrow \mathbb{Z}$ that adds coordinates.
2. Choose paths $\sigma_i : |\Delta^1| \rightarrow X$ from x_0 to x_i for $1 \leq i < k$. Prove that their relative homology classes form a basis of $H_1(X, X')$. Prove that these classes are independent of the chosen paths with the same endpoints.
3. Suppose $f : X \rightarrow X$ maps X' to itself, with $f(x_i) = x_{\pi(i)}$ for a permutation π . Describe the induced action on the kernel above and give its matrix in the basis from part 2. Recover the sign of reflection in the preceding exercise when $k = 2$.